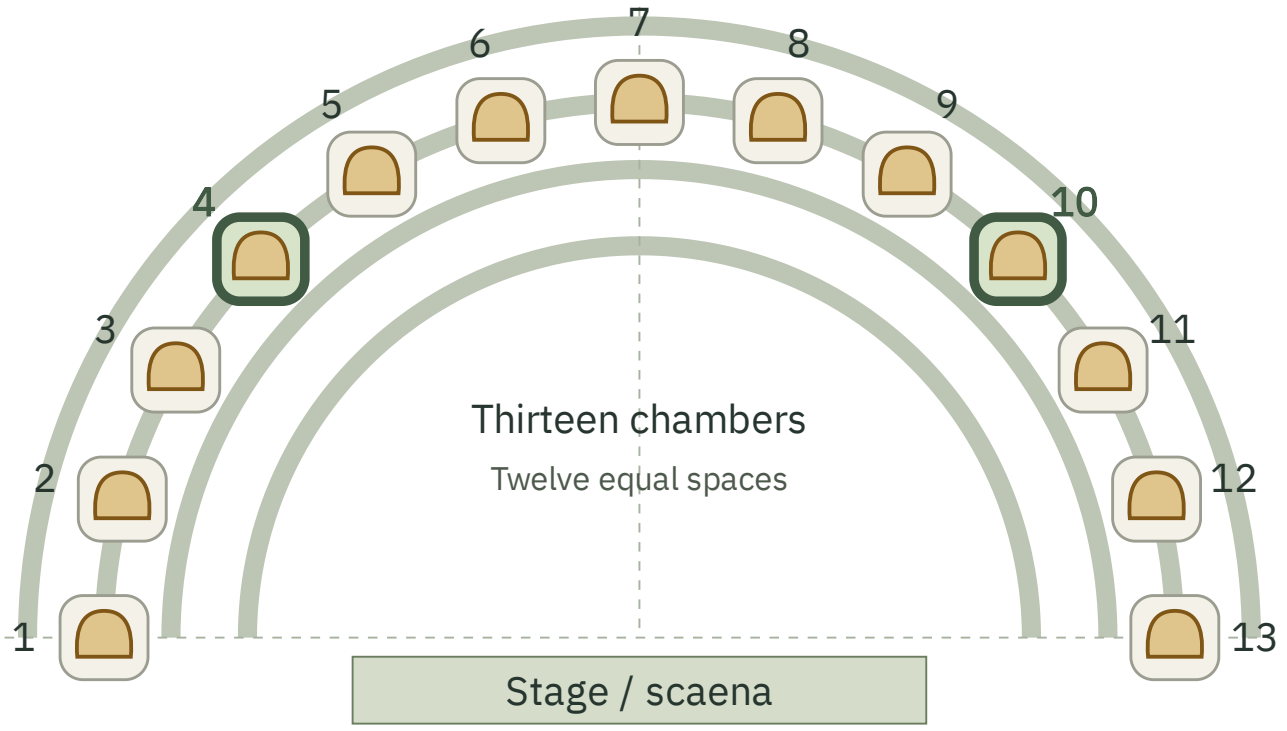
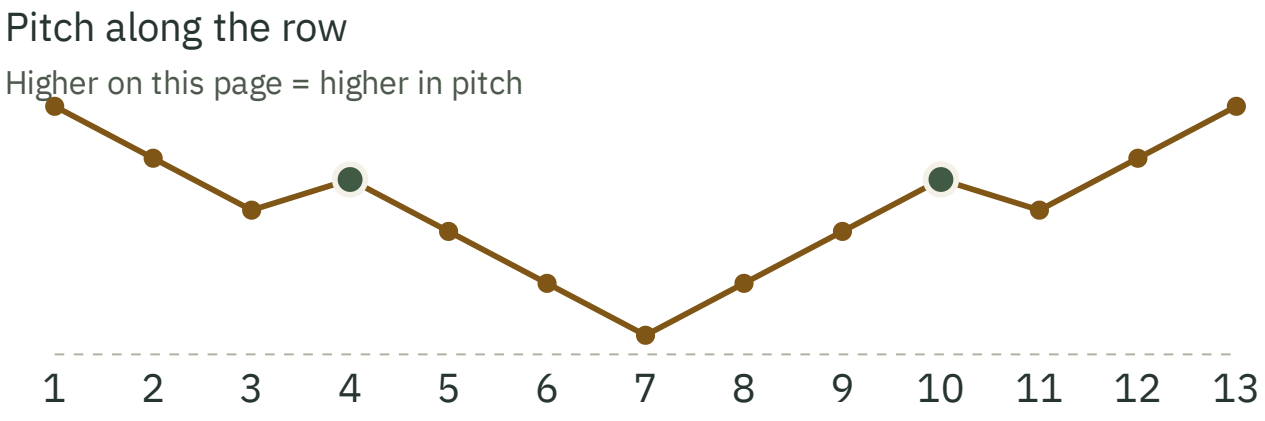
Thirteen vessels. Seven notes.

Saved arrangement: **the printed prescription**. Mese = 220 Hz, a modern listening reference.

Reopen these settings. This document embeds its drawings, audio, fonts and all five source pages. It needs no network connection.



Selected: Nete synhemmenon. The two halves mirror one another; only the central pitch has one vessel. Schematic plan, with illustrative bowl shapes.



The vertical scale represents logarithmic pitch height, not seating height.

The fourth place is not another fourth

From either end, the prescription descends from nete hyperbolaeon to nete diezeugmenon to paramese. The fourth place then rises to nete synhemmenon. Quarta counts a position; diatessaron names a musical interval. Swapping the third and fourth placements produces an ascending half-row, but alters the text.

Seven pitches; mese = 220 Hz (chosen). Places count left to right in the drawing.

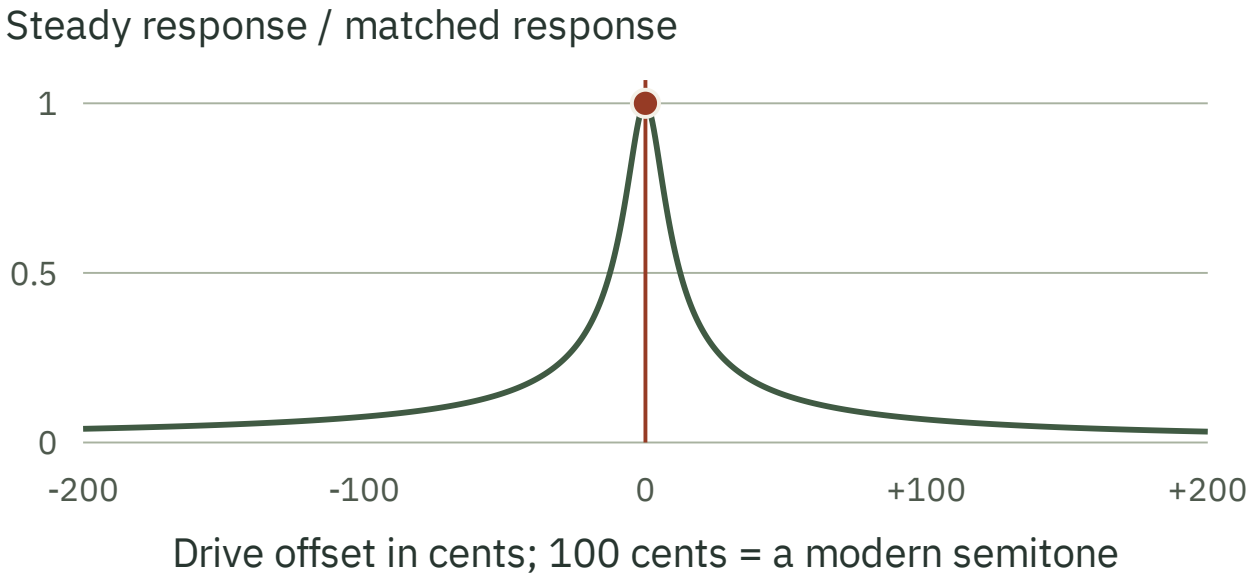
Pitch	Places	Ratio to mese	Hz
Hypate hypaton	7	9:16	123.750
Hypate meson	6, 8	3:4	165.000
Mese	5, 9	1:1	220.000
Nete synhemmenon	4, 10	4:3	293.333
Paramese	3, 11	9:8	247.500
Nete diezeugmenon	2, 12	3:2	330.000
Nete hyperbolaeon	1, 13	2:1	440.000

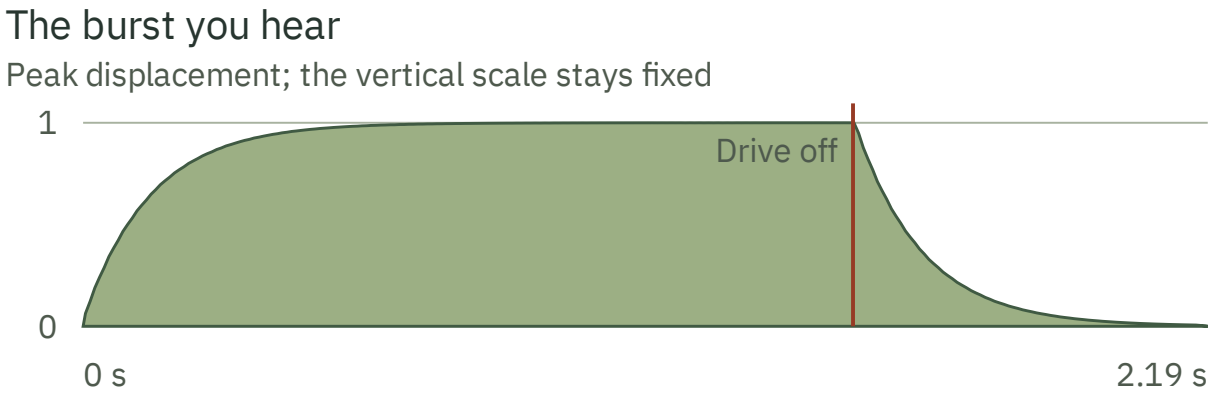
A tetrachord is a group of four notes spanning a fourth. In this model, the conjunct tetrachord runs from mese to nete synhemmenon; the disjunct one starts a tone higher, at paramese, and runs to nete diezeugmenon. Its starting note is still below the other tetrachord's top note. That overlap explains the turn in the row.

These are pure-ratio realizations of the fixed-note relationships, with fourth = 4:3, octave = 2:1 and the mese–paramese tone = 9:8. Vitruvius gives names and intervals, not these hertz values or an ancient pitch standard. The Greek forms are normalized musical names corresponding to his Latin transliterations.

One mode responds selectively

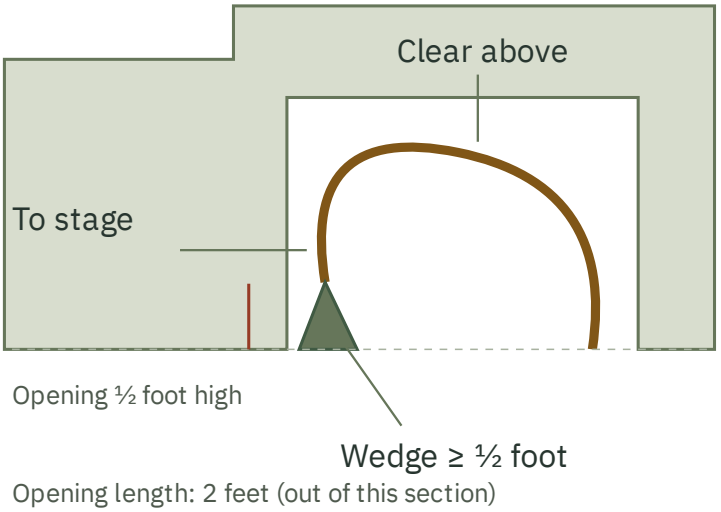
The selected mode has natural frequency 293.333 Hz. Drive: 293.333 Hz (0 cents offset); Q = 120. Steady response relative to a matched drive: 100.00%. Free decay time to 1/e amplitude: 0.1302 seconds.





The equation is $x'' + (\omega_0/Q)x' + \omega_0^2 x = (\omega_0^2/Q)\sin(\omega t)$. The drive lasts 1.5 seconds, starting from rest; the same oscillator then decays freely. x is dimensionless displacement, not pressure or loudness. Its exact solution is converted to audio at a fixed gain, so detuned examples remain quieter. The final 25 ms fades to silence.

What rests inside the chamber?



The prescribed support leaves the vessel free of the walls. Its size and exact profile are not supplied. Source feet describe the wedges and openings; they are not modern metres.

Read the evidence

Nine selected passages from F. Krohn’s 1912 edition. Line-end hyphens are joined. Angle brackets preserve editorial additions; square brackets preserve deletions. English translations and reading notes are new.

V.4.5 · pp. 102–103

1. The fixed notes

stantes autem sunt, qui inter mobiles [sunt] interpositi continent tetrachordi coniunctionem et e generum discriminibus suis finibus sunt permanentes; appellantur autem sic: proslambanomenos, hypate hypaton, hypate meson, mese, nete synhemmenon, paramese, nete diezeugmenon, nete hyperbolaeon.

The fixed notes, placed between the moving ones, maintain the connection of the tetrachord and stay at their own limits through the differences of the genera. They are named as follows: proslambanomenos, hypate hypaton, hypate meson, mese, nete synhemmenon, paramese, nete diezeugmenon, nete hyperbolaeon.

stantes / mobiles · fixed / moving

These terms describe a note’s position when changing musical genus. The one-row vessel prescription uses seven of these eight fixed notes, leaving out proslambanomenos. The square brackets preserve Krohn’s deletion of sunt. The list itself is not in ascending pitch order.

V.5.1 · p. 104

2. Tune the bronze

Ita ex his indagationibus mathematicis rationibus fiant vasa aerea pro ratione magnitudinis theatri, eaque ita fabricentur, ut, cum tangantur, sonitum facere possint inter se diatessaron, diapente ⟨et⟩ ex ordine ad disdiapason.

Following these investigations, let bronze vessels be made by mathematical rules, in proportion to the size of the theatre. Let them be fashioned so that, when struck, they can sound with one another a fourth, a fifth, and in sequence up to a double octave.

cum tangantur · when they are touched or struck

Tangantur is present passive subjunctive. This is a test of the fashioned vessels' sounding pitches; it does not supply a frequency in hertz or a vessel profile. The angle brackets mark the editor's supplied et. The general double-octave instruction does not mean the seven-note row spans two octaves: it omits the lowest fixed note.

V.5.1 · p. 104

3. Leave the vessel free

postea inter sedes theatri constitutis cellis ratione musica ibi conlocentur ita, uti nullum parietem tangant circaque habeant locum vacuum et ab summo capite spatium, ponanturque inversa et habeant in parte, quae spectat ad scaenam, suppositos cuneos ne minus altos semipede; contraque eas cellas relinquantur aperturae inferiorum graduum cubilibus longae pedes duo, altae semipede.

Then, after chambers have been made among the theatre seats, let them be placed there according to musical rules, touching no wall and having empty space around them and above their tops. Let them be set upside down, with wedges at least half a foot high beneath the side facing the stage. Opposite these chambers, let openings be left in the beds of the lower steps, two feet long and half a foot high.

ponanturque inversa · and let them be placed upside down

Inversa agrees with the neuter plural vasa, not with cellae. The foot dimensions govern wedges and openings. They do not give a bowl's diameter. The cutaway is a reading aid with an illustrative bowl shape; there is no conversion to a modern metre dimension.

V.5.2 · p. 104

4. Thirteen chambers, twelve intervals

si non erit ampla magnitudine theatrum, media altitudinis transversa regio designetur et in ea tredecim cellae duodecim aequalibus intervallis distantes confornicentur

If the theatre is not large, let a horizontal band be marked halfway up its height, and in it let thirteen vaulted chambers be built, separated by twelve equal intervals.

tredecim / duodecim · thirteen / twelve

These intervals are spaces between chambers. Thirteen positions need twelve gaps. They are not twelve equal musical intervals. The schematic plan uses equal angular gaps on a semicircle; the text locates the row halfway up the seating.

V.5.2 · pp. 104–105

5. From either end: three descending notes

uti ea echea, quae supra scripta sunt, ad neten hyperbolaeon sonantia in cellis, quae sunt in cornibus extremis, utraque parte prima conlocentur, secunda ab extremis diatessaron ad neten diezeugmenon, tertia diatessaron ad paramesen

Let the sounding vessels mentioned above, sounding nete hyperbolaeon, be placed first on each side in the chambers at the extreme wings; the second from the ends a fourth away, at nete diezeugmenon; the third a fourth away, at paramese.

utraque parte · on each side

Both halves receive the same pitches. Prima, secunda and tertia count placements inward from each end, not pitches upward from the centre. Krohn’s apparatus records corrected readings for diezeugmenon and paramesen; the model follows the printed text, not a claim that every manuscript agrees.

V.5.2 · p. 105

6. The fourth place interrupts the descent

quarta ad neten synhemmenon, quinta diatessaron ad mesen, sexta diatessaron ad hypaten meson, in medio unum diatessaron ad hypaten hypaton.

The fourth at nete synhemmenon; the fifth a fourth away, at mese; the sixth a fourth away, at hypate meson; and in the middle one vessel a fourth away, at hypate hypaton.

quarta / diatessaron · fourth placement / interval of a fourth

Quarta counts the fourth vessel from an end. There is no diatessaron before nete synhemmenon. That pitch is higher than paramese. In the chosen pure-ratio realization, the rise is 32:27, about 294 cents. Sorting by pitch swaps the third and fourth placements on each side and changes the prescription.

V.5.3 · p. 105

7. What Vitruvius expects to happen

ita hac ratiocinatione vox a scaena uti ab centro profusa se circumagens tactuque feriens singulorum vasorum cava excitaverit auctam claritatem et concentu convenientem sibi consonantiam.

By this arrangement, the voice pouring out from the stage as from a centre, spreading around and striking the hollows of the individual vessels, will bring about increased clarity and a consonance agreeing with itself through the combined sound.

auctam claritatem · increased clarity

This is the author’s proposed effect. A musical placement diagram cannot establish whether a built theatre became louder or clearer. The modern oscillator below isolates frequency selectivity; it has no seats, audience, reflections or sound-pressure calibration.

V.5.8 · p. 106

8. What he can point to

sin autem quaeritur, in quo theatro ea sint facta, Romae non possumus ostendere, sed in Italiae regionibus et in pluribus Graecorum civitatibus. etiamque auctorem habemus Lucium Mummium, qui diruto theatro Corinthiorum ea aenea Romam deportavit et de manubiis ad aedem Lunae dedicavit.

If it is asked in which theatre these things have been made, we cannot point to one at Rome, but can in regions of Italy and in several Greek cities. We also have Lucius Mummius as a witness: after destroying the Corinthians' theatre he carried its bronze vessels to Rome and made a dedication from the spoils at the temple of Luna.

Romae non possumus ostendere · we cannot point to one at Rome

The author distinguishes local examples from reported ones. This passage is literary testimony, not a surviving set of thirteen measured vessels or a record of their sound.

V.5.8 · p. 106

9. A less expensive material

multi etiam sollertes architecti, qui in oppidis non magnis theatra constituerunt, propter inopiam fictilibus doliis ita sonantibus electis hac ratiocinatione compositis perfecerunt utilissimos effectus.

Many skilful architects who built theatres in smaller towns, because of limited means, selected earthenware jars that sounded in this way and arranged them by this reasoning, achieving very useful effects.

fictilibus doliis … electis · earthenware jars having been selected

The wording gives a material alternative and a requirement to select sounding jars. It does not establish that air-cavity resonance and vibration of a bronze shell are the same mechanism. Their relative roles remain a question for historical and acoustic research.

Keep the Greek names

ὑπάτη ὑπάτων · Hypate hypaton: lowest note of the lowest tetrachord.

ὑπάτη μέσων · Hypate meson: lowest note of the middle tetrachord.

μέση · Mese: the middle note.

νήτη συνημμένων · Nete synhemmenon: highest note of the conjunct tetrachord.

παραμέση · Paramese: the note beside mese.

νήτη διεζευγμένων · Nete diezeugmenon: highest note of the disjunct tetrachord.

νήτη ὑπερβολαίων · Nete hyperbolaeon: highest note of the highest tetrachord.

The Greek forms explain the musical terms; they are not a Greek original of Vitruvius's Latin instructions.

What remains unknown

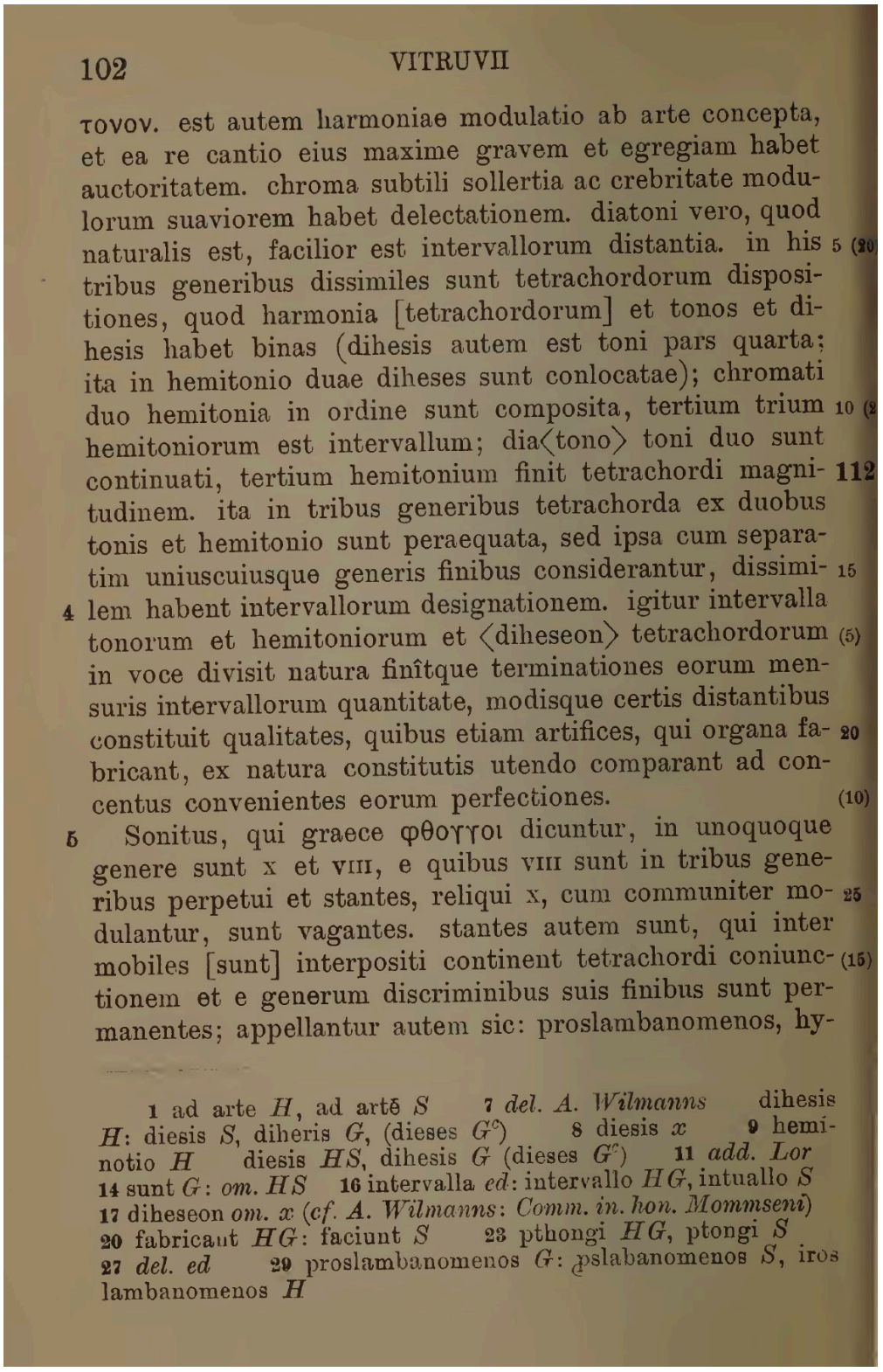
The placement follows Vitruvius V.5.2 in Krohn's edition. The semicircle is a schematic row, not a surveyed theatre. Frequencies, pure ratios, sine tones and damping are chosen modern realizations. Neither the drawing nor the single-mode experiment establishes ancient vessel shape, timbre, loudness or acoustic benefit.

Shell vibration and resonance of enclosed air are different mechanisms. A selected musical frequency alone does not identify the mechanism or predict the effect at an audience seat. Modern researchers have considered both reinforcement and absorption; their models are not a measurement of Vitruvius's vessels.

Further reading: J. H. Rindel, Echo Problems in Ancient Theatres (2011); Polychronopoulos et al., The Use of Resonators in Ancient Greek Theaters (2011). The workshop reproduces none of their figures or simulations.

Krohn, p. 102: Fixed and moving notes

V.4.3–5. The fixed-note list begins at the bottom and continues on p. 103. Source volume.



τονον. est autem harmoniae modulatio ab arte concepta, et ea re cantio eius maxime gravem et egregiam habet auctoritatem. chroma subtili sollertia ac crebritate modulorum suaviorem habet delectationem. diatoni vero, quod naturalis est, facilior est intervallorum distantia. in his tribus generibus dissimiles sunt tetrachordorum dispositiones, quod harmonia [tetrachordorum] et tonos et dihesis habet binas (dihesis autem est toni pars quarta; ita in hemitonio duae diheses sunt conlocatae); chromati duo hemitonia in ordine sunt composita, tertium trium hemitoniorum est intervallum; dia⟨tono⟩ toni duo sunt continuati, tertium hemitonium finit tetrachordi magnitudinem. ita in tribus generibus tetrachorda ex duobus tonis et hemitonio sunt peraequata, sed ipsa cum separatim uniuscuiusque generis finibus considerantur, dissimi-
4 lem habent intervallorum designationem. igitur intervalla tonorum et hemitoniorum et ⟨diheseon⟩ tetrachordorum in voce divisit natura finîtque terminationes eorum mensuris intervallorum quantitate, modisque certis distantibus constituit qualitates, quibus etiam artifices, qui organa fabricant, ex natura constitutis utendo comparant ad concentus convenientes eorum perfectiones.

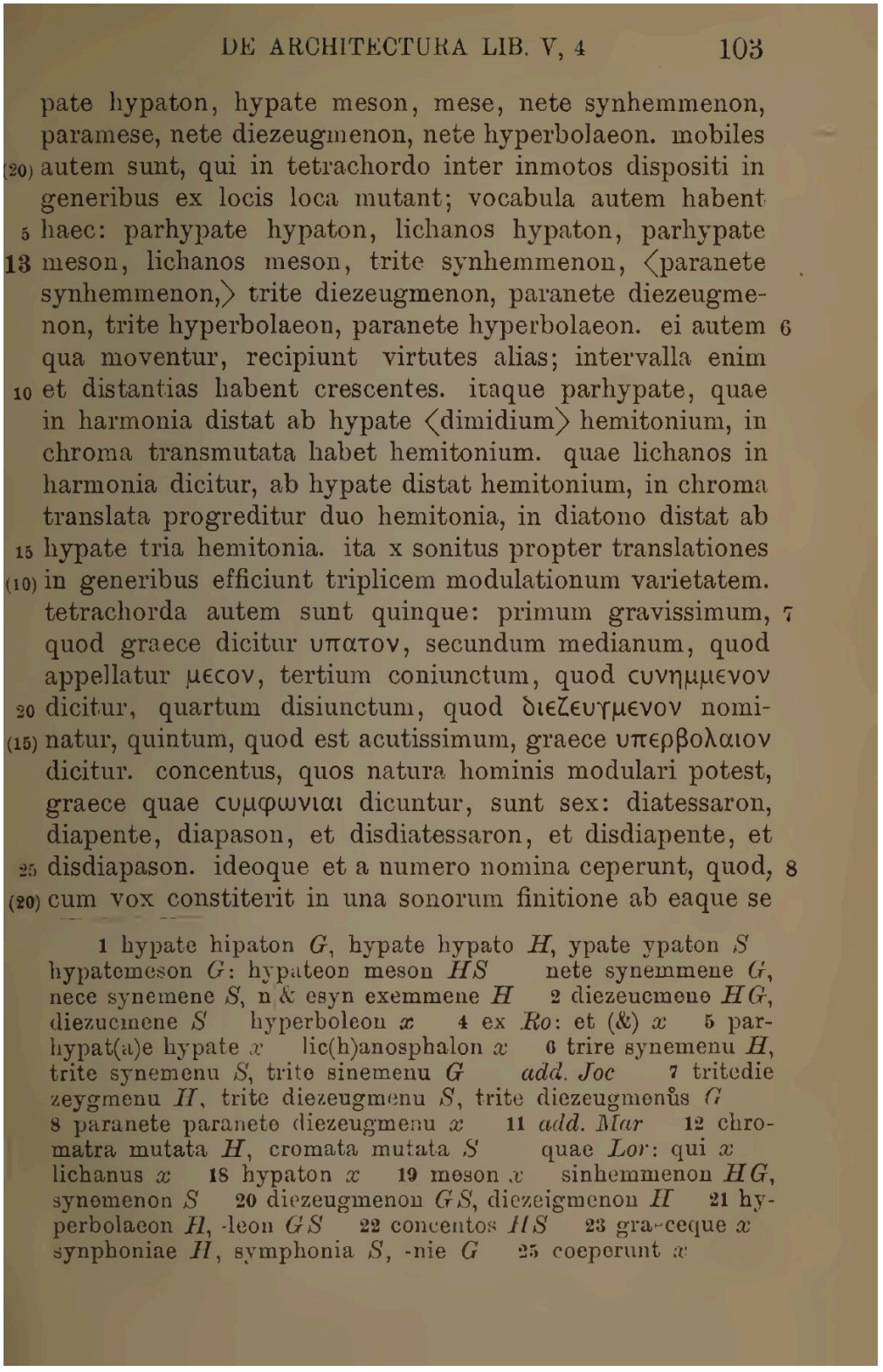
5 Sonitus, qui graece φθογγοι dicuntur, in unoquoque genere sunt X et VIII, e quibus VIII sunt in tribus generibus perpetui et stantes, reliqui X, cum communiter modulantur, sunt vagantes. stantes autem sunt, qui inter mobiles [sunt] interpositi continent tetrachordi coniunctionem et e generum discriminibus suis finibus sunt permanentes; appellantur autem sic: proslambanomenos, hy-

1 ad arte *H*, ad artē *S* 7 *del. A. Wilmanns* dihesis *H*: diesis *S*, diheris *G*, (dieses G^c) 8 diesis *x* 9 heminotio *H* diesis *HS*, dihesis *G* (dieses G^c) 11 *add. Lor* 14 sunt *G*: *om. HS* 16 intervalla *ed*: intervallo *HG*, intuallo *S* 17 diheseon *om. x* (*cf. A. Wilmanns*: *Comm. in. hon. Mommseni*) 20 fabricant *HG*: faciunt *S* 23 pthongi *HG*, ptongi *S* 27 *del. ed* 29 proslambanomenos *G*: ꝓslabanomenos *S*, iros lambanomenos *H*

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Krohn, p. 103: The names of the fixed notes

V.4.5–8. The first two lines finish the list. The lower discussion distinguishes conjunct and disjunct tetrachords. Source volume.



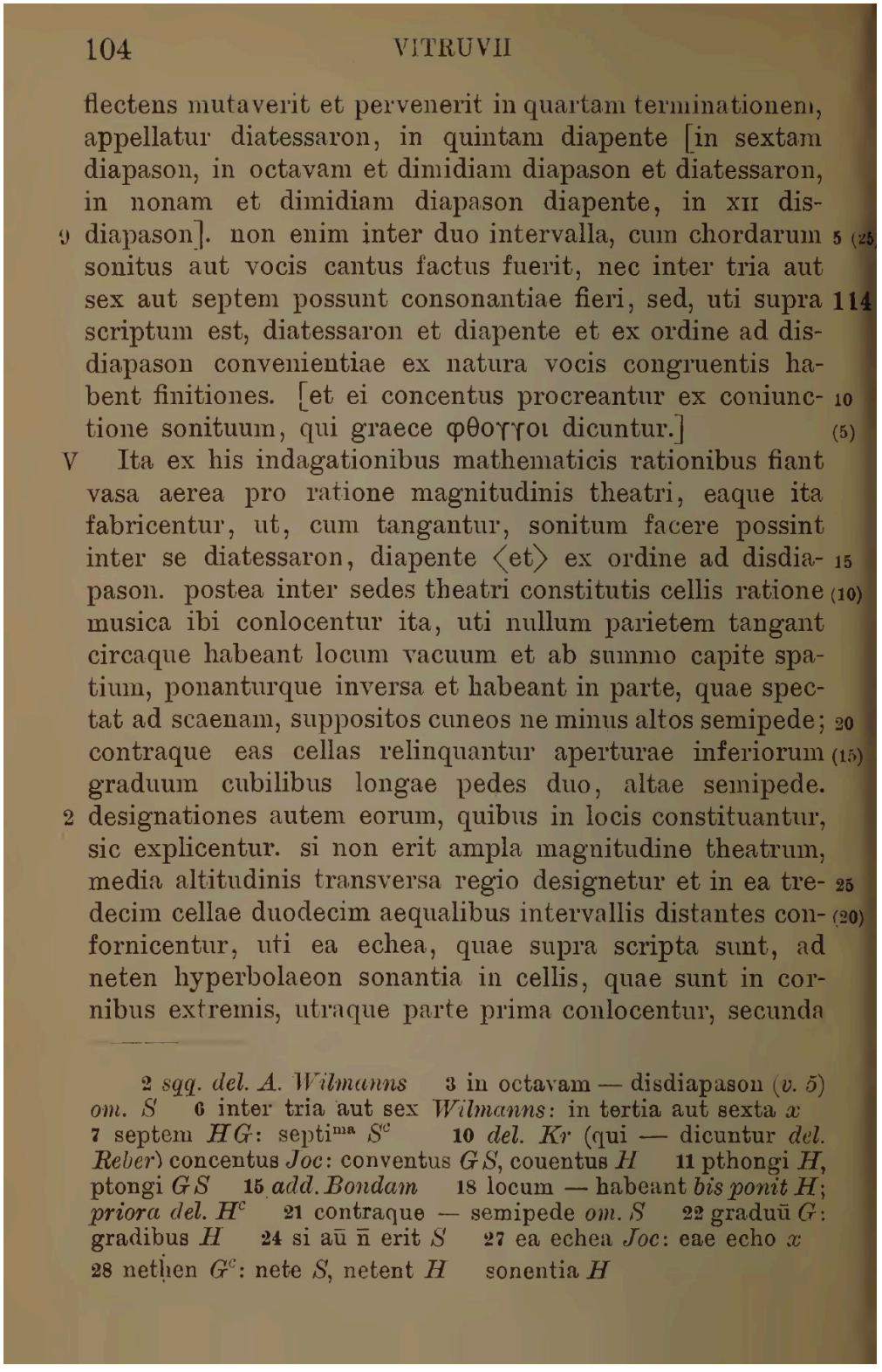
pate hypaton, hypate meson, mese, nete synhemmenon,
paramese, nete diezeugmenon, nete hyperbolaeon. mobiles
(20) autem sunt, qui in tetrachordo inter inmotos dispositi in
generibus ex locis loca mutant; vocabula autem habent
5 haec: parhypate hypaton, lichanos hypaton, parhypate
13 meson, lichanos meson, trite synhemmenon, ⟨paranete
synhemmenon,⟩ trite diezeugmenon, paranete diezeugme-
non, trite hyperbolaeon, paranete hyperbolaeon. ei autem 6
qua moventur, recipiunt virtutes alias; intervalla enim
10 et distantias habent crescentes. itaque parhypate, quae
in harmonia distat ab hypate ⟨dimidium⟩ hemitonium, in
chroma transmutata habet hemitonium. quae lichanos in
harmonia dicitur, ab hypate distat hemitonium, in chroma
translata progreditur duo hemitonia, in diatono distat ab
15 hypate tria hemitonia. ita x sonitus propter translationes
(10) in generibus efficiunt triplicem modulationum varietatem.
tetrachorda autem sunt quinque: primum gravissimum, 7
quod graece dicitur υπατον, secundum medianum, quod
appellatur μεcον, tertium coniunctum, quod cυνημμενον
20 dicitur, quartum disiunctum, quod διεζευγμενον nomi-
(15) natur, quintum, quod est acutissimum, graece υπερβολαιον
dicitur. concentus, quos natura hominis modulari potest,
graece quae cυμφωνιαι dicuntur, sunt sex: diatessaron,
diapente, diapason, et disdiatessaron, et disdiapente, et
25 disdiapason. ideoque et a numero nomina ceperunt, quod, 8
(20) cum vox constiterit in una sonorum finitione ab eaque se

1 hypate hipaton *G*, hypate hypato *H*, ypate ypaton *S* hypatemeson *G*: hypateon meson *HS* nete synemmene *G*, nece synemene *S*, n & esyn exemmene *H* 2 diezeucmene *HG*, diezucmene *S* hyperboleon *x* 4 ex *Ro*: et (&) *x* 5 parhypat(a)e hypate *x* lic(h)anosphalon *x* 6 trire synemenu *H*, trite synemenu *S*, trite sinemenu *G* *add. Joc* 7 tritedie zeygmenu *H*, trite diezeugmenu *S*, trite diezeugmenûs *G* 8 paranete paranete diezeugmenu *x* 11 *add. Mar* 12 chromatra mutata *H*, cromata mutata *S* quae *Lor*: qui *x* lichanus *x* 18 hypaton *x* 19 meson *x* sinhemmenon *HG*, synemenon *S* 20 diezeugmenon *GS*, diezeigmenon *H* 21 hyperbolaeon *H*, -leon *GS* 22 concentos *HS* 23 gra-ceque *x* synphoniae *H*, symphonia *S*, -nie *G* 25 coeperunt *x*

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Krohn, p. 104: Make, support and place the vessels

V.5.1–2 begins halfway down. The angle-bracketed et is Krohn's editorial supplement. Read the full page and apparatus. Source volume.



104 VITRUVII

flectens mutaverit et pervenerit in quartam terminationem, appellatur diatessaron, in quintam diapente [in sextam diapason, in octavam et dimidiam diapason et diatessaron, in nonam et dimidiam diapason diapente, in XII dis-
9 diapason]. non enim inter duo intervalla, cum chordarum sonitus aut vocis cantus factus fuerit, nec inter tria aut sex aut septem possunt consonantiae fieri, sed, uti supra scriptum est, diatessaron et diapente et ex ordine ad disdiapason convenientiae ex natura vocis congruentis habent finitiones. [et ei concentus procreantur ex coniunctione sonituum, qui graece φθογγοι dicuntur.]

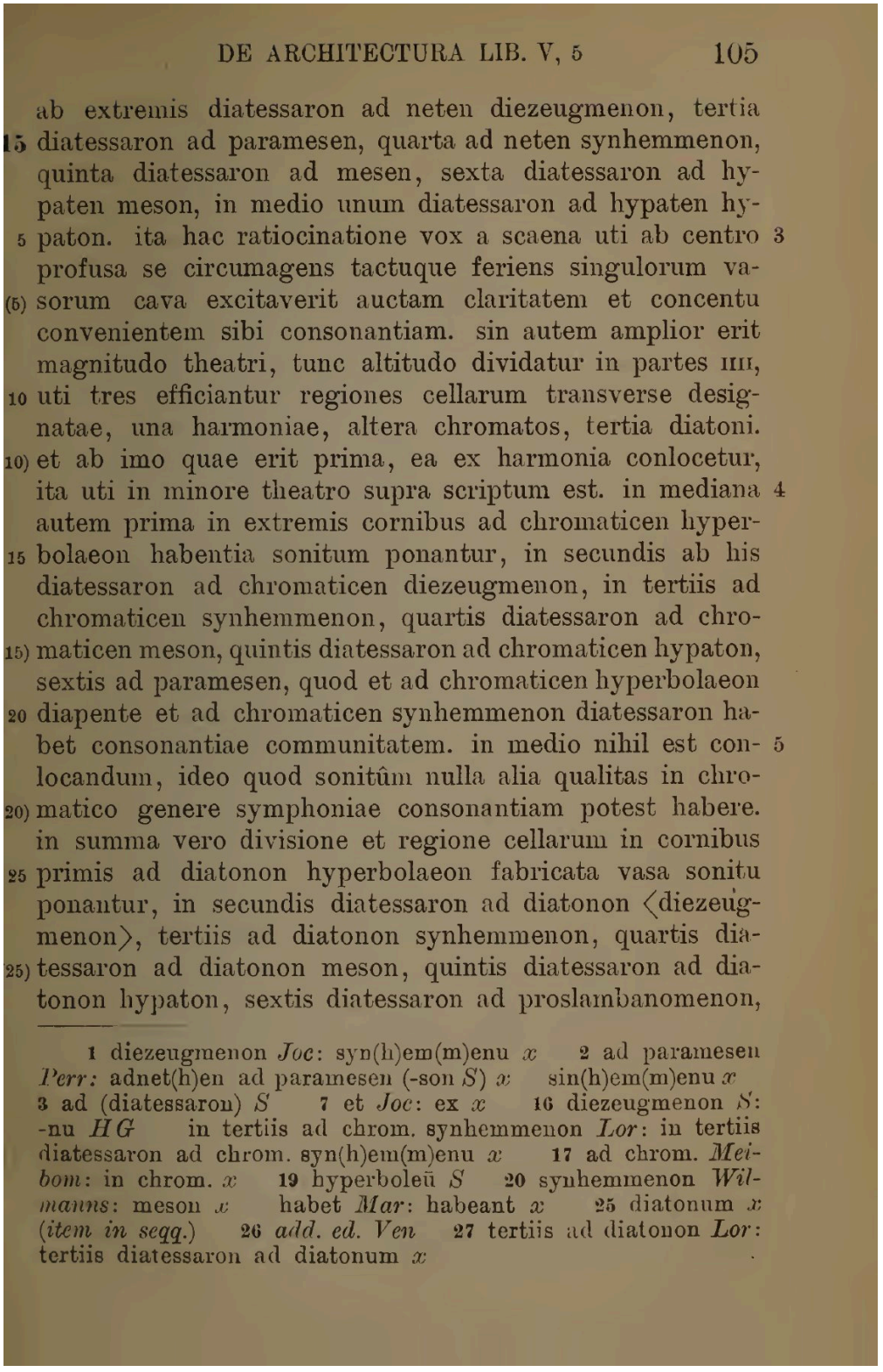
V Ita ex his indagationibus mathematicis rationibus fiant vasa aerea pro ratione magnitudinis theatri, eaque ita fabricentur, ut, cum tangantur, sonitum facere possint inter se diatessaron, diapente ⟨et⟩ ex ordine ad disdiapason. postea inter sedes theatri constitutis cellis ratione musica ibi conlocentur ita, uti nullum parietem tangant circaque habeant locum vacuum et ab summo capite spatium, ponanturque inversa et habeant in parte, quae spectat ad scaenam, suppositos cuneos ne minus altos semipede; contraque eas cellas relinquantur aperturae inferiorum graduum cubilibus longae pedes duo, altae semipede.
2 designationes autem eorum, quibus in locis constituantur, sic explicentur. si non erit ampla magnitudine theatrum, media altitudinis transversa regio designetur et in ea tredecim cellae duodecim aequalibus intervallis distantes confornicentur, uti ea echea, quae supra scripta sunt, ad neten hyperbolaeon sonantia in cellis, quae sunt in cornibus extremis, utraque parte prima conlocentur, secunda

2 *sqq. del. A. Wilmanns* 3 in octavam — disdiapason (*v.* 5) *om. S* 6 inter tria aut sex *Wilmanns*: in tertia aut sexta *x* 7 septem *HG*: septi^{ma} S^{c} 10 *del. Kr* (qui — dicuntur *del. Reber*) concentus *Joc*: conventus *GS*, couentus *H* 11 pthongi *H*, ptongi *GS* 15 *add. Bondam* 18 locum — habeant *bis ponit H*; *priora del.* H^{c} 21 contraque — semipede *om. S* 22 graduū *G*: gradibus *H* 24 si aū n̄ erit *S* 27 ea echea *Joc*: eae echo *x* 28 nethen G^{c}: nete *S*, netent *H* sonentia *H*

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Krohn, p. 105: The row, its intended effect, and larger theatres

V.5.2 continues at the top. The larger theatre's three rows appear below; this workshop models only the one-row prescription. Source volume.



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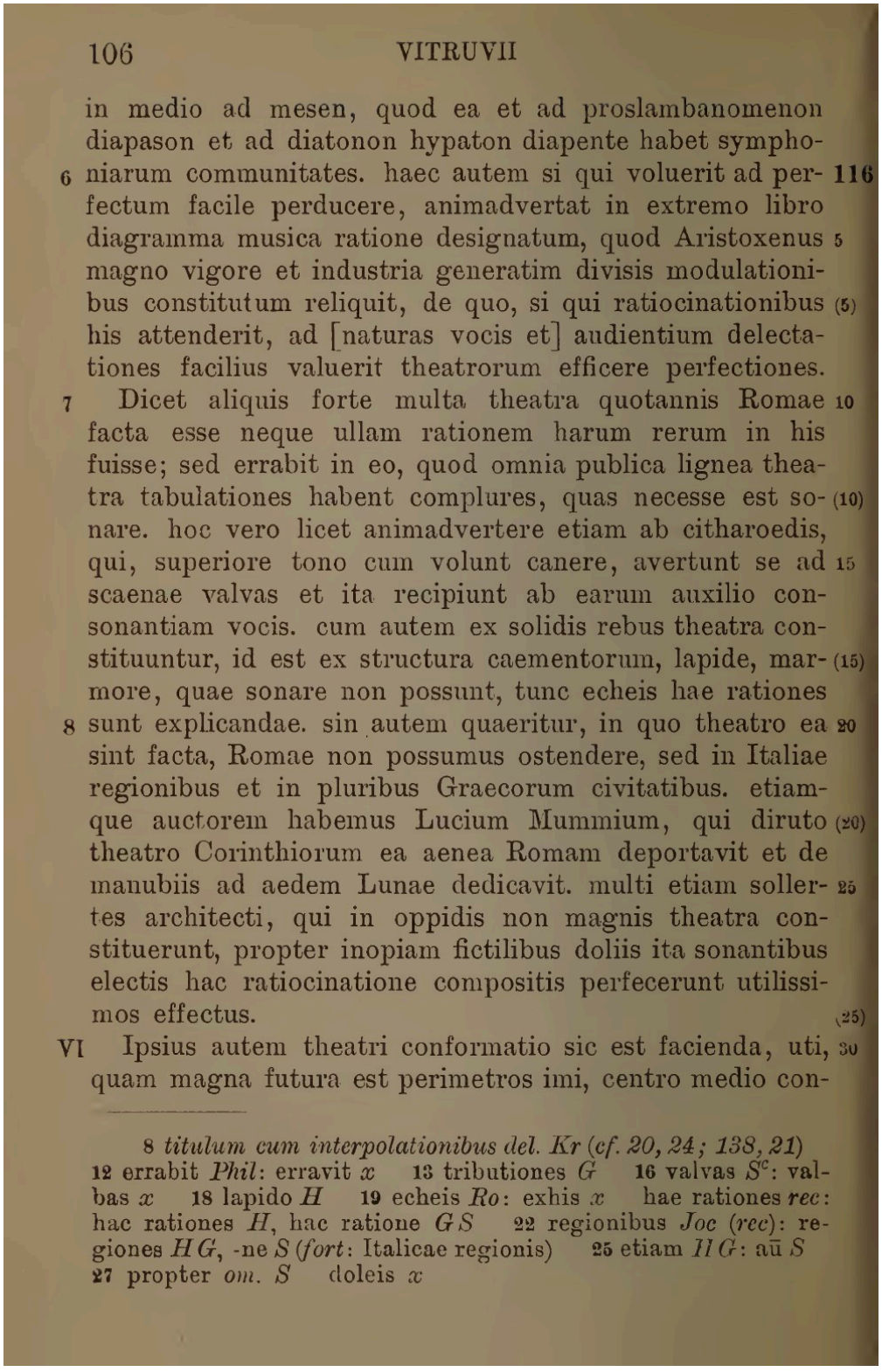
ab extremis diatessaron ad neten diezeugmenon, tertia diatessaron ad paramesen, quarta ad neten synhemmenon, quinta diatessaron ad mesen, sexta diatessaron ad hypaten meson, in medio unum diatessaron ad hypaten hypaton. ita hac ratiocinatione vox a scaena uti ab centro 3 profusa se circumagens tactuque feriens singulorum vasorum cava excitaverit auctam claritatem et concentu convenientem sibi consonantiam. sin autem amplior erit magnitudo theatri, tunc altitudo dividatur in partes IIII, uti tres efficiantur regiones cellarum transverse designatae, una harmoniae, altera chromatos, tertia diatoni. et ab imo quae erit prima, ea ex harmonia conlocetur, ita uti in minore theatro supra scriptum est. in mediana 4 autem prima in extremis cornibus ad chromaticen hyperbolaeon habentia sonitum ponantur, in secundis ab his diatessaron ad chromaticen diezeugmenon, in tertiis ad chromaticen synhemmenon, quartis diatessaron ad chromaticen meson, quintis diatessaron ad chromaticen hypaton, sextis ad paramesen, quod et ad chromaticen hyperbolaeon diapente et ad chromaticen synhemmenon diatessaron habet consonantiae communitatem. in medio nihil est con- 5 locandum, ideo quod sonitûm nulla alia qualitas in chromatico genere symphoniae consonantiam potest habere. in summa vero divisione et regione cellarum in cornibus primis ad diatonon hyperbolaeon fabricata vasa sonitu ponantur, in secundis diatessaron ad diatonon ⟨diezeugmenon⟩, tertiis ad diatonon synhemmenon, quartis diatessaron ad diatonon meson, quintis diatessaron ad diatonon hypaton, sextis diatessaron ad proslambanomenon,

1 diezeugmenon *Joc*: syn(h)em(m)enu *x* 2 ad paramesen *Perr*: adnet(h)en ad paramesen (-son *S*) *x* sin(h)em(m)enu *x* 3 ad (diatessaron) *S* 7 et *Joc*: ex *x* 16 diezeugmenon *S*: -nu *HG* in tertiis ad chrom. synhemmenon *Lor*: in tertiis diatessaron ad chrom. syn(h)em(m)enu *x* 17 ad chrom. *Meibom*: in chrom. *x* 19 hyperboleũ *S* 20 synhemmenon *Wilmanns*: meson *x* habet *Mar*: habeant *x* 25 diatonum *x* (*item in seqq.*) 26 *add. ed. Ven* 27 tertiis ad diatonon *Lor*: tertiis diatessaron ad diatonum *x*

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Krohn, p. 106: The limits of the author’s examples

V.5.6–8. Vitruvius cannot point to an example at Rome. He reports Greek and Italian examples, Mummius’s spoils, and cheaper earthenware alternatives. Source volume.



106 VITRUVII

in medio ad mesen, quod ea et ad proslambanomenon diapason et ad diatonon hypaton diapente habet symphoniarum communitates. haec autem si qui voluerit ad perfectum facile perducere, animadvertat in extremo libro diagramma musica ratione designatum, quod Aristoxenus magno vigore et industria generatim divisis modulationibus constitutum reliquit, de quo, si qui ratiocinationibus his attenderit, ad [naturas vocis et] audientium delectationes facilius valuerit theatrorum efficere perfectiones.

7 Dicet aliquis forte multa theatra quotannis Romae facta esse neque ullam rationem harum rerum in his fuisse; sed errabit in eo, quod omnia publica lignea theatra tabulationes habent complures, quas necesse est sonare. hoc vero licet animadvertere etiam ab citharoedis, qui, superiore tono cum volunt canere, avertunt se ad scaenae valvas et ita recipiunt ab earum auxilio consonantiam vocis. cum autem ex solidis rebus theatra constituuntur, id est ex structura caementorum, lapide, marmore, quae sonare non possunt, tunc echeis hae rationes

8 sunt explicandae. sin autem quaeritur, in quo theatro ea sint facta, Romae non possumus ostendere, sed in Italiae regionibus et in pluribus Graecorum civitatibus. etiamque auctorem habemus Lucium Mummium, qui diruto theatro Corinthiorum ea aenea Romam deportavit et de manubiis ad aedem Lunae dedicavit. multi etiam sollertes architecti, qui in oppidis non magnis theatra constituerunt, propter inopiam fictilibus doliis ita sonantibus electis hac ratiocinatione compositis perfecerunt utilissimos effectus.

VI Ipsius autem theatri conformatio sic est facienda, uti, quam magna futura est perimetros imi, centro medio con-

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8 *titulum cum interpolationibus del. Kr (cf. 20, 24; 138, 21)*
12 errabit *Phil*: erravit *x* 13 tributiones *G* 16 valvas S^c: valbas *x* 18 lapido *H* 19 echeis *Ro*: exhis *x* hae rationes *rec*: hac rationes *H*, hac ratione *G S* 22 regionibus *Joc (rec)*: regiones *H G*, -ne *S (fort*: Italicae regionis) 25 etiam *H G*: aū *S*
27 propter *om. S* doleis *x*

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